

EXP 4

A* ALGORITHM IMPLEMENTATION

```
from queue import PriorityQueue

def a_star(graph, start, goal, heuristic):
    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}

    # Initialize g-scores
    g_score = {node: float('inf') for node in graph}
    g_score[start] = 0

    while not open_set.empty():
        _, current = open_set.get()

        if current == goal:
            path = []

            while current in came_from:
                path.append(current)
                current = came_from[current]

            path.append(start)

            return path[::-1]

        for neighbor, cost in graph[current]:
            tentative_g = g_score[current] + cost

            if tentative_g < g_score[neighbor]:
                came_from[neighbor] = current
                g_score[neighbor] = tentative_g

                # f(n) = g(n) + h(n)
                f = tentative_g + heuristic[neighbor]

                open_set.put((f, neighbor))

    return None

graph = {
    'A': [('B', 1), ('C', 4)],
```

```
'B': [('D', 2), ('E', 5)],  
'C': [('F', 1)],  
'D': [('G', 2)],  
'E': [('G', 1)],  
'F': [('G', 3)],  
'G': []  
}  
  
heuristic = {  
    'A': 7,  
    'B': 6,  
    'C': 2,  
    'D': 1,  
    'E': 0,  
    'F': 3,  
    'G': 0  
}  
  
path = a_star(graph, 'A', 'G', heuristic)  
  
print("Shortest path using A*:", path)
```

PS E:\poai> python A_star_algo.py

Shortest path using A*: ['A', 'B', 'D', 'G']